

Forecasting the Supreme Court with Honest Features: Walk-Forward Vote Prediction, Coalition-Aware Aggregation, and a Live Two-Stage Preregistered Registry

Kara Zajac 

Independent researcher kara@soulstone.org

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Abstract

We present JUDGMENT, an end-to-end system for forecasting Supreme Court behavior at the level of individual justice votes and case outcomes, built on the Supreme Court Database (SCDB) and evaluated exclusively by walk-forward validation over 69 terms (1956–2024). A gradient-boosted model restricted to strictly pre-decision features attains **67.9% justice-vote accuracy** (Brier 0.207) and **69.4% case-outcome accuracy** in its deployed cert-stage configuration — exceeding our own full-SCDB research configuration and decisively beating base-rate, justice-history, and attitudinal baselines (case-clustered bootstrap, +3.3 points over the strongest baseline, 95% CI [+2.7, +3.8]). A separately validated **post-argument stage** adds per-justice questioning features parsed from 7,023 oral-argument transcripts and the side the Solicitor General supports as amicus, reaching **69.6% justice-vote accuracy** (Brier 0.200, AUC 0.723); the SG signal alone is worth +5.2 points where it applies, and the combination resolves a format regime change that had erased questioning signal in the post-2020 seriatim era. Four methodological contributions travel beyond this system. First, *deployment honesty*: the configuration that publishes forecasts is restricted to features actually knowable for a pending case and is validated separately; the unrestricted model produces degenerate forecasts under deployment missingness. Second, a *two-factor coalition model* over calibrated per-justice marginals replaces the ubiquitous vote-independence assumption, improving out-of-sample split-distribution log-loss by 45% while also improving binary outcome accuracy. Third, an *audited adoption gate*: every candidate feature faces the same walk-forward test, five rejections are documented alongside four adoptions, and a nested-selection replay shows the entire adoption sequence reproduces itself when decided on pre-2011 evidence alone — scoring 71.5% justice-vote accuracy on the untouched 2011–2024 window. Fourth, a *live two-stage preregistered registry*: cert-stage forecasts freeze on argument day, post-argument forecasts re-register separately, and a strictly ex-ante scoring rule — enforced in code, exercised by its first real case — makes post-hoc revision both detectable and inert.

Keywords: Supreme Court, judicial behavior, forecasting, walk-forward validation, calibration, oral argument, preregistration.

*Code, data, and every validation artifact: <https://github.com/KaraZajac/JUDGMENT>; archived at doi:10.5281/zenodo.21211376. Live forecasts: <https://judgment.karazajac.io>.

1 Introduction

Quantitative prediction of Supreme Court behavior has a distinguished lineage: ideal-point models of judicial ideology (Martin and Quinn, 2002), pre-confirmation ideology scores (Segal and Cover, 1989), the statistical-models-versus-experts contest of the 2002 term (Ruger et al., 2004), and general machine-learning models spanning two centuries of decisions (Katz et al., 2017a). Yet published systems are difficult to audit *as forecasters*: evaluations vary in leakage discipline, probability calibration is rarely reported, case-level probabilities are almost universally computed under an indefensible independence assumption across the nine votes, and — most fundamentally — retrospective accuracy does not establish that a system’s *deployable* configuration, facing the information actually available before decision, performs as advertised.

This paper describes a system built to be auditable in exactly those respects, and reports what such discipline reveals. Our contributions:

1. **A leak-free evaluation harness** (§4): walk-forward only — train on terms $\leq T-1$, predict every vote of term T — with an explicit feature-availability policy and prospective recalibration.
2. **Deployment honesty** (§4.6): the published forecaster is a separately validated configuration restricted to stage-appropriate features. We document that the full research model, whose accuracy headlines would normally be quoted, collapses to degenerate output under deployment missingness — a failure mode invisible to standard evaluation.
3. **Coalition-aware aggregation** (§4.5): a two-factor probit structure over calibrated marginals that recovers the Court’s unanimity modes without sacrificing — indeed while improving — binary outcome accuracy.
4. **An audited feature-adoption gate** (§5.2): four adoptions and five documented rejections under one protocol, including a twice-tested negative result for question text, the measurement of the Solicitor-General-as-amicus effect, and the discovery and resolution of a format regime change in oral argument.
5. **A methods audit** (§5.3): the three sharpest criticisms we could level at our own procedure — adaptive selection on a reused validation window, clustered dependence in significance tests, and an unvalidated attribution heuristic — each converted into an experiment and answered.
6. **A live two-stage preregistered registry** (§6): forecasts for every granted-but-undecided case, timestamped in a public git history, with cert-stage forecasts frozen at argument, post-argument forecasts scored as their own track, and a strictly ex-ante scoring rule enforced in code.

2 Related work

Ideal points. Martin and Quinn (2002) introduced dynamic Bayesian item-response modeling of justices’ latent positions; their scores are the field’s standard ideology measure. We re-estimate ideal points in a penalized-ML variant of their model rather than consuming published scores, attach parametric-bootstrap uncertainty, and test their *predictive* contribution — which proves nil beyond lagged behavioral rates, an instructive redundancy.

Prediction. Ruger et al. (2004) showed simple classification trees beating a panel of experts on the 2002 term (75% vs. 59% case accuracy). Katz et al. (2017a) built time-evolving random forests over SCDB features for 1816–2015, reporting 70.2% case and 71.9% justice-vote accuracy out-of-sample; theirs is the benchmark general model, but two regime differences make the comparison indicative rather than head-to-head: their feature discipline is decision-date knowability rather than cert-stage (features include whether oral argument was heard, whether a rehearing occurred, and the elapsed time from argument to decision), and their margin over the always-petitioner rule is roughly two points (Medvedeva et al., 2023, whose survey characterizes it as “a small improvement over the 68% baseline”). Kaufman et al. (2019) report the highest published academic figure — 74.04% case accuracy over 2005–2015 — but the entire above-baseline margin comes from oral-argument features: their case-covariates-only model scores 60.6%, *below* the 67.98% always-petitioner baseline, under pooled ten-fold cross-validation rather than walk-forward evaluation. Read against our results (§5.2), that finding independently corroborates this paper’s central negative result — at the cert stage, structure is nearly all there is, and predictive content arrives with argument.

Crowds and markets. The FantasySCOTUS crowd is the one system that genuinely outperforms our numbers, and it does so on later information: over OT2011–2016 (425 cases, 7,284 participants) fair aggregation rules score $\sim 74.1\%$ case / 73.5% justice-vote, with predictions revisable until the eve of decision and no probabilities, Brier, or calibration reported (Katz et al., 2017b); the widely quoted 80.8% is the authors’ own post-hoc ceiling across 277,201 aggregation configurations. Prediction markets provide no systematic docket coverage. No verified LLM-based system reports accuracy above ours (Hamilton, 2023, 60% case); the JUSTICE text benchmark reports $\leq 68\%$ despite augmented case variants straddling a random train/test split (Alali et al., 2021); and Medvedeva et al. (2023) document that much of the wider “legal judgment prediction” literature classifies post-decision documents rather than forecasting from pre-decision information. As of mid-2026, no algorithmic system publishes live, calibrated, preregistered forecasts of pending cases; the registry of §6 occupies an empty lane.

Oral argument. That questioning at argument reveals the likely outcome is well established: the side receiving more questions tends to lose (Johnson et al., 2009; Epstein et al., 2010), and Kaufman et al. (2019) built their above-baseline margin on argument text. Our contribution is to measure the signal per-justice, side-attributed, across seven decades under walk-forward discipline — and to document a format regime change that compresses it (§5.2).

3 Data

3.1 Canonical corpus

The Supreme Court Database (Spaeth et al., 2025) supplies 29,202 cases and 255,832 justice-vote records spanning the 1791–2024 terms. Our pipeline restructures the corpus into per-case records (votes embedded, closed code sets decoded against the SCDB codebook) and per-justice records combining curated biography with computed voting histories. Structural validation — identifier coherence, vote-token validity, tally consistency against SCDB’s own counts, membership integrity — runs on every build; residual soft findings are upstream SCDB quirks and are documented rather than silently repaired. Justice biographies for the modern era are hand-curated

from public record; earlier justices are enriched from the Federal Judicial Center’s Biographical Directory.

3.2 The living edge

SCDB is released annually, so the term in progress is invisible to the canonical dataset for months. Three ingestion layers close the gap, each writing records flagged provisional and replaced wholesale when SCDB coverage arrives: Oyez (Oyez Project, 2026) supplies decided-case metadata and (with days-to-weeks lag) per-justice votes; CourtListener (Free Law Project, 2026) supplies near-real-time decision events; and the Court’s own slip-opinion syllabi are parsed for same-day per-justice lineups. The syllabus parser is *validation-gated*: it must reproduce the lineups of independently coded cases before it may write anything (it currently agrees on 43 of 44 such cases, 98%). One subtlety matters for correctness: every upstream source keys consolidated decisions to the lead docket alone, so a map parsed from the syllabus’s “Together with No. ...” footnote resolves companion dockets to their deciding record — seven such pairs in OT2025 alone.

3.3 Auxiliary corpora

Question-presented texts for 3,418 historical cases support the text experiments of §5.2. An **oral-argument questioning corpus** covers 7,023 argued cases — 96% of the 1955–2024 terms, never below 94% in any decade — parsed from Oyez’s structured per-speaker transcripts into per-justice, side-attributed turn and word counts. Sides resolve from advocate descriptions where present; in the pre-1970s metadata gap they resolve from argument order (petitioner opens and rebuts), a heuristic validated blind at 99.0% section agreement on the two-advocate structure that dominates the era it serves (§5.3). The same advocate metadata yields the side the United States supports as amicus at argument (723 cases). Lower-court opinions matched via docket-number joins accumulate as training data for a direction classifier (in progress; the deployed system uses hand-coded direction with documented bases).

4 Methods

4.1 Targets

Primary: does justice j vote to **disturb the judgment below** (reverse family = reversed / vacated / partial reversals; affirm = affirmed), defined where SCDB codes both disposition and majority membership — the observable, ideology-coding-free target comparable to Katz et al. (2017a). Secondary: the SCDB liberal/conservative **direction** of the vote, which inherits the Spaeth coding conventions and their critiques. Case outcome is the majority side of the (up to) nine votes.

4.2 Features and the availability policy

Every feature must be knowable *before* the decision, and the deployed configurations restrict further to features knowable at their stage:

- **Case features (cert stage)**: hand-codable issue area, jurisdiction, U.S.-as-party flags, and the ideological direction of the decision below — the anchor for the latter being that

certiorari petitioners lost below, classified under SCDB conventions, with ambiguous cases left honestly uncoded.

- **Justice features:** computed from votes in terms strictly before the row’s term (term-granular expanding windows): tenure, prior reverse rate, prior liberal share (career and last-3-terms, and within issue area), prior majority/dissent rates — all empirical-Bayes shrunk toward running strictly-prior means — plus the appointing president’s party. First-term justices carry pure priors; the cold start is honest.
- **Argument features (post-argument stage only):** per-justice words and turns directed at each side while it held the floor; the justice’s word share toward the petitioner minus her own lagged three-term mean share (so style and format norms cancel); bench-wide aggregates; and the side the Solicitor General supports as amicus.

Nothing downstream of the vote (disposition, direction, majority size, opinion data) is ever a feature. We follow the literature — and flag the optimism — in treating SCDB’s post-hoc issue coding as cert-stage-knowable for historical evaluation; the deployed forecaster uses hand-coded provisional issue areas instead, and §5.3 pre-commits to measuring the coding agreement when SCDB’s release covers the forecast terms.

4.3 Walk-forward protocol

No random splits, ever: random splits leak court composition and era effects. For every evaluation term $T \in [1956, 2024]$ we train a gradient-boosted classifier (histogram-based, native categorical and missing-value handling (Pedregosa et al., 2011)) on all labeled votes from terms $\leq T-1$ and predict every vote of term T . Baselines are computed strictly from the same training window: the training base rate; the justice’s lagged shrunk prior rate; and the classic attitudinal heuristic, $P(\text{reverse}) = P(\text{justice disagrees with the ideological direction of the decision below})$. Features whose history begins mid-window (e.g., the SG signal, sparse before the 1960s) simply sit out training windows in which they are entirely missing.

4.4 Prospective recalibration

Raw model probabilities are recalibrated by isotonic regression fitted, for each evaluation term T , only on out-of-sample predictions from terms before T — exactly what a live forecaster could have done at the time. This reduces vote-level expected calibration error to ≈ 0.02 – 0.03 with slight Brier improvements; Figure 2 shows the resulting reliability. The deployed calibrators are exported as portable step functions, so any environment reproduces calibrated probabilities bit-exactly.

4.5 Coalition-aware aggregation

Aggregating nine per-justice probabilities under independence is indefensible: it predicts 4.4% unanimous reversals in a Court that is unanimous about a third of the time. We place a two-factor probit coalition structure on top of the calibrated marginals, preserving them exactly:

$$\text{vote}_j \mid (u, v) \sim \text{Bernoulli}(\Phi(\mu_j + \lambda_0 u + \lambda_1 s_j v)), \quad \mu_j = \sqrt{1 + \lambda_0^2 + \lambda_1^2 s_j^2} \Phi^{-1}(p_j),$$

	Accuracy	Brier	Log loss	AUC
<i>Baselines (reverse target)</i>				
Training base rate	.646	.229	—	.487
Justice history (lagged)	.646	.227	—	.548
Attitudinal heuristic	.623	.228	—	.664
<i>Models (reverse target)</i>				
Bare cert subset	.644	.224	.640	.601
+ lower-court direction	.678	.208	.603	.686
+ recent topic lean (deployed cert stage)	.679	.207	.602	.688
Post-argument stage (deployed)	.696	.200	.587	.723
<i>Direction target (liberal)</i>				
Deployed cert stage	.661	.211	.610	.727
Post-argument stage	.680	.207	.603	.744

Table 1: Walk-forward vote-level performance, evaluation terms 1956–2024 ($n = 75,787$ reverse-labeled votes; 78,874 direction-labeled). Case-level accuracy of the deployed cert configuration under coalition aggregation is 69.4% ($n = 7,645$ cases). Every model row is trained, predicted, and calibrated strictly out-of-time.

where u is a case-valence factor, v an ideological factor, s_j the justice’s lagged lean, and the loadings (λ_0, λ_1) are fit on walk-forward predictions from terms ≤ 2010 and evaluated strictly after. Case distributions follow from two-dimensional Gauss–Hermite quadrature; no Monte Carlo. Out-of-sample split-distribution log-loss improves by 45%, and — non-obviously — binary case accuracy improves as well (69.0% $\rightarrow$ 69.4%): it is strictly better, not a sharpness trade (Figure 3).

4.6 Deployment honesty

The configuration that publishes forecasts is **pinned** — never inferred from artifact presence — and is validated end-to-end like every other configuration, with its own out-of-sample predictions supplying its calibrator. This discipline is not decorative. When the full research model was first pointed at pending cases, its never-seen missingness constellation (absent lower-court codings, party codes, court codes) routed into unreliable leaves and produced degenerate ≈ 1.00 forecasts for every case — a catastrophic failure invisible to conventional evaluation, caught only because deployment was treated as its own validated configuration. The deployed cert-stage subset in fact *outperforms* the full research configuration (67.9% vs. 66.8% justice-vote), a lean-model result that recurs throughout §5.2.

5 Results

5.1 Headline performance

Table 1 reports walk-forward vote-level performance over 1956–2024; Figure 1 shows the time path. The deployed cert-stage configuration beats the strongest baseline by +3.3 points (case-clustered 95% CI [+2.7, +3.8]; §5.3), and the post-argument configuration adds a further +1.7 points overall — much more where its information applies (§5.2).

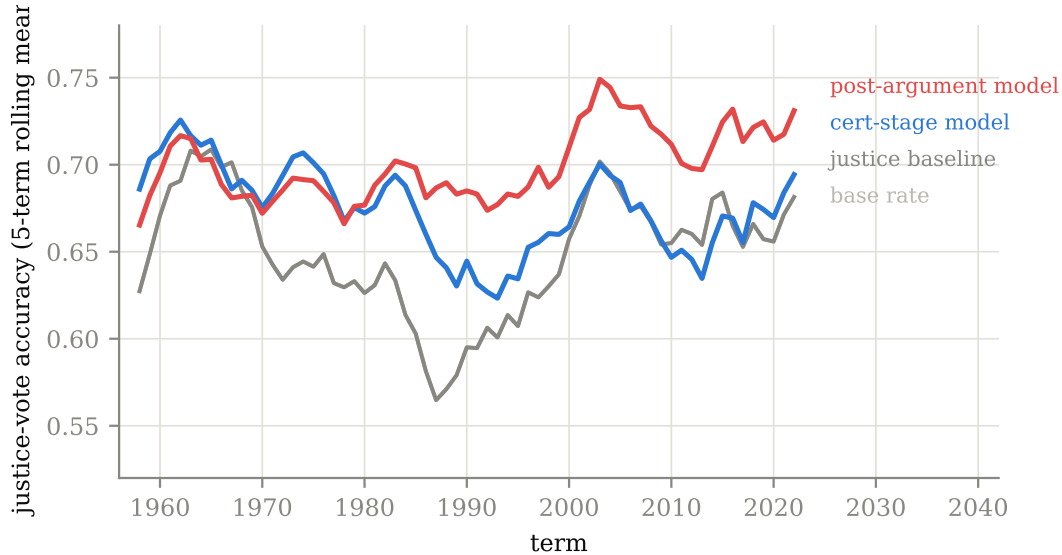


Figure 1: Justice-vote accuracy by term (5-term rolling mean), walk-forward. The cert-stage model’s margin over the justice-history baseline is stable across seven decades; the post-argument model’s additional margin widens in the modern era as transcript coverage and Solicitor General participation grow.

5.2 The feature-adoption gate

Every candidate feature faces the same protocol — walk-forward evaluation on both targets, judged on accuracy, Brier, and AUC — and the ledger records losers alongside winners (Table 2). Three findings deserve narrative.

Structure beats content at the cert stage. Question-presented text, entered through leakage-safe per-step latent semantic features, does not improve prediction either alone or — refuting the interaction hypothesis directly — in the presence of lower-court direction. Meanwhile one hand-codable structural judgment (the direction of the decision below) is worth 3.4 points. Three further cert-stage candidates — a nomination-time ideology score, the originating circuit, and the presence of a dissent below — all failed the same gate, the circuit categorical reproducing in miniature the variance pathology that sank the full research configuration. Five straight rejections of features that re-encode information already implicit in the lean configuration constitute evidence of a *cert-stage information ceiling* near 68% per-vote for this model class.

The questioning signal and a format regime change. Per-justice questioning features add +1.73 points on transcript-covered votes, with an era structure that is itself a finding (Figure 4): +3.8 to +4.7 points sustained across the 1980s–2010s — the per-justice, walk-forward counterpart of Kaufman et al. (2019)’s case-level result — negative in the 1950s–60s (cold benches; heuristic attribution), and initially *zero* in the 2020s. The cause is mechanical: since OT2020 the Court appends seriatim justice-by-justice questioning rounds to free-for-all argument, compressing turn-count asymmetries. Word-volume features with a within-justice lagged baseline (so format and style norms cancel), combined with the SG signal below, lift the 2020s covered slice from 68.6%

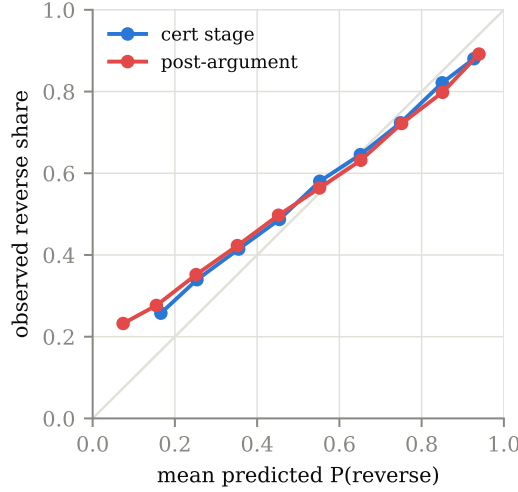


Figure 2: Reliability of the two deployed stages (decile bins, all evaluation terms pooled). Prospective isotonic recalibration holds both stages close to the diagonal.

Candidate	Stage	Verdict	Evidence (reverse target)
Lower-court direction	cert	adopted	+3.45pp [+3.02, +3.86]
Recent topic lean	cert	adopted	Brier/AUC; accuracy null
Question text (LSA), twice	cert	rejected	−0.5pp alone; no lc interaction
Segal–Cover nomination score	cert	rejected	null; first-term +0.7pp underpowered
Originating circuit	cert	rejected	worse on all six metrics
Lower-court dissent	cert	rejected	flat-to-worse both targets
Questioning (turns), oral	post-arg.	adopted [†]	+1.73pp [+1.11, +2.33] covered
Questioning (word-centric)	post-arg.	adopted	Brier everywhere; seriatim-robust
SG-as-amicus side	post-arg.	adopted	+5.19pp [+3.32, +7.10] covered
Union (turns + words + SG)	post-arg.	rejected	worse direction-target profile

Table 2: The adoption ledger. Brackets are case-clustered bootstrap 95% confidence intervals (§5.3). [†]Turn-based questioning was adopted and then superseded by the word-centric + SG configuration. Five rejections and four adoptions under one protocol; the file-drawer is empty.

to 72.5% (Brier 0.2045 → 0.1872): the current Court’s format becomes the model’s best modern era rather than its weakest.

The Solicitor General effect, measured. The side the United States supports as amicus at argument is the strongest single feature we have tested: +5.19 points on covered votes (95% CI [+3.32, +7.10]). The mechanism is the classic one (Epstein et al., 2013), here measured within our own corpus: justices vote to reverse 75.8% of the time when the SG backs the petitioner and 42.3% when it backs the respondent (Figure 5), and the model prices the swing almost perfectly (73.6% / 44.7% predicted). Because merits amicus participation is not knowable at cert, the feature belongs to the post-argument stage — whose transcript harvest is also, conveniently, its data source.

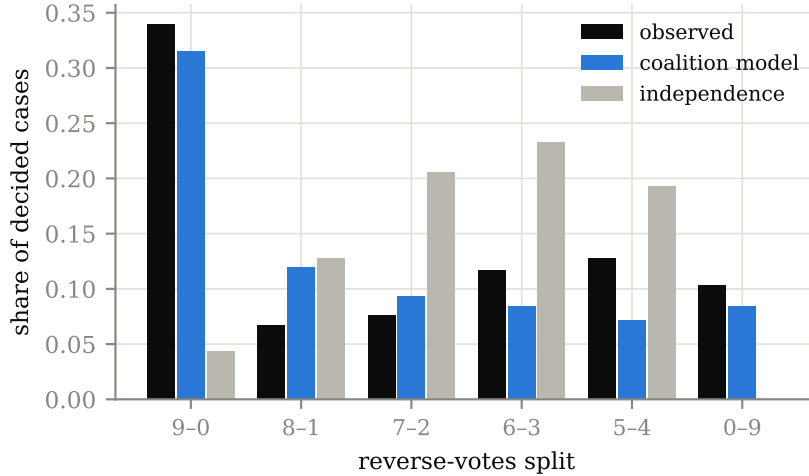


Figure 3: Distribution of reverse-vote splits: observed, coalition model, and independence aggregation of identical calibrated marginals. Independence places 4.4% on unanimous reversal against an observed 34.0%; the two-factor coalition model recovers both unanimity modes. Residual honesty: 5–4 mass remains underpredicted.

Configuration (chosen on ≤ 2010)	Reverse			Liberal		
	Acc	Brier	AUC	Acc	Brier	AUC
Bare cert subset	.645	.230	.534	.596	.239	.632
Deployed cert stage	.663	.217	.623	.645	.221	.700
Deployed post-argument	.715	.190	.737	.690	.202	.754

Table 3: Nested-selection holdout: every adoption decision made on 1956–2010 evidence only; performance reported on 2011–2024, which no decision touched ($n = 8,224$ reverse-labeled votes).

5.3 A methods audit

Three criticisms of our own procedure were converted into experiments.

Adaptive selection. Every gate above reuses the same 1956–2024 window; adopting the best of ten comparisons risks overfitting the validation set. A *nested-selection replay* re-makes every adopt/reject decision from metrics computed on 1956–2010 alone, under the same decision rules. The nested procedure reproduces the deployed configuration chain *exactly* — every rejection, every adoption, the final post-argument configuration winning its tournament six metrics to zero — and the surviving model scores **71.5% justice-vote accuracy (Brier 0.190, AUC 0.737)** on the untouched 2011–2024 window (Table 3): better than its full-window average. The selection procedure did not manufacture the headline.

Clustered inference. Naive per-vote tests overstate evidence because nine votes share a case. All load-bearing comparisons were re-tested with a case-clustered bootstrap (4,000 draws); the intervals in Table 2 are the result. Every accuracy-based adoption survives; the one null (topic lean) matches its documented probability-quality adoption basis, and the nested replay independently re-adopts it.

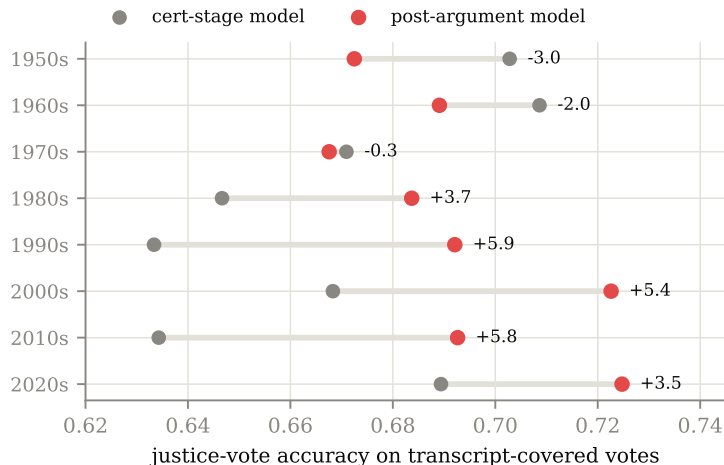


Figure 4: Justice-vote accuracy on transcript-covered votes by decade: deployed cert-stage model vs. the deployed post-argument model. The post-2020 seriatim format initially erased the questioning signal; word-centric features with a within-justice baseline plus the SG signal make the 2020s the strongest modern era.

The attribution heuristic, validated blind. Pre-1970s transcripts lack advocate metadata, so sides resolve from argument order. On 527 description-era cases (1,760 sections, 61,955 justice-turns) with descriptions deleted, the heuristic agrees with ground truth on **99.0%** of sections in two-advocate arguments — the structure that dominates the era it serves — degrading to 61% only in the multi-advocate arguments that, in the description era, do not need it. This quantifies the early-era noise visible in Figure 4.

6 The live registry

Every granted-but-undecided case receives a forecast: calibrated $P(\text{reverse})$, per-justice probabilities for both targets, and the full coalition split distribution, each file carrying model version, feature-availability notes, and hand-coding bases. Registration is **staged**. Cert-stage forecasts are re-issued as hand-codings improve and *freeze on argument day* — the registered cert record cannot absorb argument-informed recoding or newer model vintages. A post-argument stage re-registers argued cases under its own timestamps once transcript features exist, and is scored as its own track, so the live record will measure what argument information is worth — including whether the seriatim-era recovery of §5.2 holds prospectively.

Every vintage is timestamped in the public git history, and the scorer enforces the preregistration property directly rather than by convention: *a forecast whose file was (re)generated on or after its case’s decision date is excluded from the track record as not ex-ante, never scored*. Post-hoc revision is thus both detectable and inert. The rule has already been exercised: the registry’s first resolution, *Little v. Hecox*, was decided as a consolidated companion of *West Virginia v. B.P.J.* days before its forecast’s final regeneration, and enters the record as excluded — although it would have scored as a hit. At this writing the registry holds forecasts for all 29 pending cases (every one leaning reverse, reflecting certiorari selection’s ~70% recent-term reversal base rate); the post-argument track begins with the October 2026 sitting. The system self-updates daily — interim ingest, same-day syllabus vote extraction, transcript harvest,

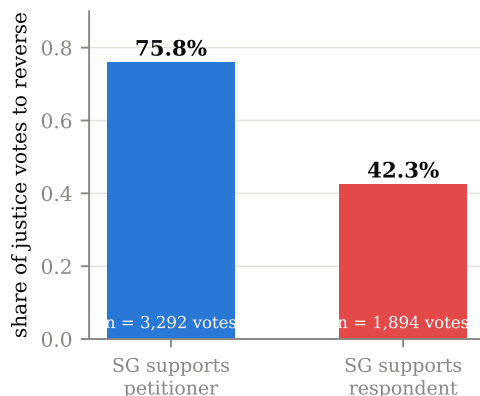


Figure 5: The Solicitor General effect in the corpus: share of justice votes to reverse conditional on the side the United States supports as amicus at argument (1955–2024; 723 cases).

scoring — with all changes committed by continuous integration.

7 Limitations

(1) Historical evaluation treats SCDB issue coding as cert-stage-knowable; deployment uses hand-coded provisional values with documented bases — an asymmetry we surface rather than hide, with a standing pre-commitment: when SCDB’s release covers the forecast terms, we publish the agreement rate between our hand-codes and SCDB’s official coding, whatever it is. (2) Lower-court direction, the most valuable cert-stage feature, is presently a human judgment call per grant (about two minutes; classifier assistance in progress) — coding bases are published for audit. (3) The coalition model underpredicts 5–4 outcomes; two factors cannot fully express bimodal polarization. (4) Forecasts are conditional on a merits ruling: mootness, dismissals as improvidently granted, and other procedural off-ramps lie outside the outcome space. (5) The direction target inherits SCDB’s ideological coding conventions and their critiques. (6) Ideal points carry bootstrap standard errors, not full posteriors. (7) Docket composition drifts across the evaluation window ($\sim 150 \rightarrow \sim 60$ cases per term); per-decade slices are reported and no single headline number should be quoted without them. (8) Provisional-era records are re-writable until SCDB supersedes them; scored outcomes flag their resolution basis. (9) The SG-as-amicus feature derives from Oyez’s post-hoc advocate metadata describing a pre-argument fact; deployment depends on that metadata posting promptly.

8 Conclusion

A Supreme Court forecaster can be simultaneously simple, honest, and strong: a lean set of behavioral priors plus one structural judgment about the decision below reaches the cert-stage information ceiling, and two argument-stage signals — who the bench presses, and whom the Solicitor General backs — carry it several points further, provided the evaluation never peeks. The methodological centerpiece is not any single accuracy number but the discipline connecting them: leakage-audited features, walk-forward-only evaluation, prospective calibration, deployment-matched validation, coalition-aware probabilities, an adoption ledger whose rejec-

tions are published, a nested-selection audit, and a public two-stage registry that will confirm or embarrass every claim here in real time.

Data and code availability

All code (MIT), the derived dataset, validation artifacts, the adoption ledger, and the forecast registry are public at <https://github.com/KaraZajac/JUDGMENT> and archived at doi:10.5281/zenodo.21211376 (Zajac, 2026); the browsable site is <https://judgment.karazajac.io>. The dataset derives from the Supreme Court Database (Spaeth et al., 2025) — the canonical source for case and vote records, used with attribution and gratitude — alongside Oyez (Oyez Project, 2026), CourtListener (Free Law Project, 2026), the Federal Judicial Center, and public-domain federal court opinions. Layered attribution and licensing are documented in `DATA-RIGHTS.md`.

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